The Chisel Fallacy: What AI Detection Misses and What the Intellectual Work Ledger Gets Right

Pangram Is a Very Sophisticated Way of Asking the Wrong Question

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For most of writing's history, textual production and intellectual production were tightly enough coupled that one could stand in for the other. Producing coherent, structured prose required thinking, and thinking reliably produced prose. That coupling has weakened. The tools that generate fluent, structured text no longer require the intellectual work that once produced it. In some domains, this debate is already settled.

The current manifestation of that breakdown is a tool called Pangram. In late July 2026, Substack integrated Pangram, giving any reader the ability to scan any post and receive a score indicating AI involvement. Writers called it a witch hunt (404 Media, 2026). Weeks later, Pangram's co-founder and CTO published a blog post arguing that AI detection works precisely because post-training constrains language models to consistent, detectable patterns (Emi, 2026).

Both responses make sense from where each side is standing. The problem is that Pangram's score identifies something real — evidence of AI involvement in the production of the text — and then that score gets used to answer a different question: whether the author supplied the intellectual contribution behind it. A detector can be entirely correct on its own target — identifying statistical evidence of model involvement — and still be insufficient to settle whether the named author supplied the intellectual contribution. Treating one as a proxy for the other is a category error. This paper calls it the Chisel Fallacy.

The paper names the assumption that makes this error structurally tempting, shows the consequences that follow from it, and proposes a better standard for evaluating intellectual work in an age when the coupling between textual production and intellectual production can no longer be assumed.

Two Things Getting Conflated

> **Definition**
>
> The Vending Machine Assumption treats near-total delegation as representative of AI-assisted writing, making detected involvement appear sufficient to establish absent intellectual contribution.

> **Definition**
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> Directed Authorship is a tool-agnostic workflow in which an author retains intellectual direction and responsibility for what an artifact claims and why, while delegating parts of its production.

Consider what one-shot AI writing actually looks like. Someone opens a browser, types a request — "write a 500-word essay arguing that standardized testing improves educational outcomes" — and receives organized prose. The essay has a thesis, supporting points, and a conclusion. The process took ninety seconds. The person who produced this document has performed little or none of the intellectual work the finished artifact appears to represent. The intellectual decisions — what to argue, how to structure it, which evidence to select, whether the reasoning holds — were largely delegated along with the prose.

This is the version of AI-assisted writing that most institutional concern is correctly aimed at. And it explains why textual AI detection feels intuitively sufficient: if your mental model is prompt → finished essay, then detecting AI involvement in the text looks almost equivalent to detecting that intellectual work was absent. This paper calls that mental model the Vending Machine Assumption — the belief that AI-assisted writing is this activity, and therefore that a score indicating AI involvement is evidence of absent intellectual contribution. It is the assumption that makes detector-based authorship judgments appear coherent. The assumption is not irrational — it reflects a mental model that makes sense for one way of using AI. AI-assisted work spans a broad agency gradient. The same category of tool use can range from near-total delegation to sustained human direction, judgment, and verification. The Vending Machine Assumption collapses that gradient by treating one endpoint as the category itself.

The Other Workflow

Directed Authorship is not a new activity — it is a pattern that has existed across professional writing for as long as execution could be delegated. The definition here makes explicit what that pattern involves.

There is a different mode of AI-assisted writing, and this paper calls it Directed Authorship. In this mode the author retains intellectual responsibility for what the artifact claims and why. The author determines the argument's structure, evaluates

which evidence matters, decides what the work needs to establish, and takes responsibility for whether the resulting reasoning holds. The model or collaborator handles the lower-level work: syntax, phrasing, spell-checking, formatting — what the programming profession, which has already navigated this split, calls syntax rather than orchestration. The author concentrates on the upper-level work: argument structure, evidence evaluation, reasoning, judgment. Directed Authorship is not specific to AI — it describes any mode of production in which intellectual direction is retained by the author while lower-level execution is delegated to any capable agent, human or otherwise.

Directed Authorship is a workflow definition, not a verdict on any particular author's intellectual contribution. Whether a specific author using this mode actually retained the command and responsibility it describes is exactly the question that requires inspection — and that a detection score cannot answer. A detector can identify evidence of AI involvement somewhere along the agency gradient. It cannot, from that evidence alone, locate the intellectual workflow on it.

What counts as lower-level work varies by domain. Fields Medal winner Terence Tao, writing on mathematics following the 2026 International Congress of Mathematicians, describes the emerging practice: automation handles computation and proof verification; humans determine what to prove and interpret what results mean (Tao, 2026). The structure holds across the domains this paper draws on:

Domain	**Lower-level work (delegatable)**	**Upper-level work (authorship)**
Academic writing	Syntax, phrasing, spell check, formatting	Argument structure, evidence selection, causal inference, judgment
Mathematics	Computation, equation solving, proof verification, symbolic manipulation	Determining what to prove, interpreting what results mean, mathematical intuition
Software	Code syntax, boilerplate, compilation	System design, architecture, what the software should do

A useful test: can this task be performed correctly without knowing what the artifact is trying to establish? If yes — formatting, spell checking, syntax — it belongs in the left column. If no — which comparison period to use, whether the inference holds, what the argument requires — it belongs in the right. The line moves depending on domain and context. The structure does not.

A detector identifies statistical evidence associated with AI involvement in the text. Research confirms that AI-assisted editing — even the guideline-compliant kind — produces detection flags at high rates (Karr et al., 2026). But the score

does not establish which intellectual workflow produced the text. Under the Vending Machine Assumption, a high score appears to settle the question. Once Directed Authorship exists as a workflow, the score is insufficient to settle it.

What a detector score cannot recover is which intellectual workflow produced the text — and that is exactly the question that requires a different standard.

The Shelf You Already Built

Consider a thought experiment. You read an article — a genuinely useful DIY idea, maybe fixing something around the house. You tried it. It worked. Three months later you learn the article was AI-assisted. Are you going to un-fix the shelf?

The question is slightly absurd, which is the point. Learning that AI helped produce the prose does not retroactively change whether the idea was sound, the instructions were clear, or the shelf worked. The question you were actually asking — does this hold up? — is a different question from where the sentences came from.

You don't need to know who Satoshi Nakamoto was to evaluate Bitcoin's white paper. The mechanism either works or it doesn't. The cryptographic assumptions can be tested, the protocol can be implemented, the incentive structure can be examined — none of that requires settling the author's identity (Nakamoto, 2008). Satoshi's identity matters for history, attribution, and questions about holdings and intent. It is not a premise in the protocol. Evaluate the mechanism, not the biography.

In many forms of practical and intellectual writing, we already accept that producing the sentences and supplying the underlying judgment need not be the same act. That separation is not novel to AI. It has been present throughout.

That observation is not a claim that origin is irrelevant. Provenance matters — for credit, for accountability, for disclosure. Those are real questions worth asking. But they are not the questions a statistical detector answers. A flag on the lower-level execution layer does not tell you who supplied the ideas, who is responsible for the claims, or whether the disclosure was appropriate.

The question the shelf was never asking was who formatted the instructions or chose the words. The question was whether the upper-level idea held up when you tried it. The Chisel Fallacy is what happens when we start answering the second question with evidence from the first.

Who Actually Holds the Chisel

> **Definition**
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> The Chisel Fallacy occurs when evidence about an artifact's production takes the place of examining its intellectual merit or its author's intellectual contribution. A detector can be accurate while the inference drawn from its score is still the fallacy.

> **Key idea**
>
> One question concerns what the work contributes. The other concerns what its author contributed.

Delegating the execution of a task does not establish that intellectual contribution was absent. When execution is handled by someone else — a collaborator, a model, an apparatus — the question of who directed what remains open. Intellectual contribution can reside above direct execution. The two are separable, and their separation is not new.

Recent philosophical work is arriving here independently. The Apt Curation Model, published in Philosophy & Technology in 2026, argues that authorship in AI-assisted work resides in higher-order decisions — structuring, evaluating, synthesizing — not in who produced the sentences (Rodrigues, 2026). The paper takes the next question seriously: if intellectual contribution does not map cleanly onto sentence production, what evidence can actually settle whether it was present?

The Chisel Fallacy names the error that occurs when we answer that question with the wrong evidence. It is the error of judging intellectual work by evidence about its lower-level execution rather than evidence about what it claims and whether those claims hold. The chisel tells you something about who executed the lower-level work — in Pangram's case, syntax production, phrasing, and word choice. It tells you nothing that settles the question of intellectual contribution. The Chisel Fallacy is not the act of noting how something was produced, or raising disclosure questions. It occurs when lower-level execution evidence is promoted into a verdict on intellectual contribution or validity — a verdict that, applied to a developer who directs code generation without writing every line, would now be immediately recognized as indefensible.

Automation lowers the execution floor faster than institutions update their definition of authorship. That lag is what makes the fallacy acute now.

The detector flags the chisel. It is completely blind to the sculptor.

The Delegation That Was Already There

Writing has always accommodated the separation of intellectual direction from lower-level execution. The professional publishing apparatus makes this explicit. A developmental editor restructures arguments and challenges reasoning. A copy editor improves sentence-level prose. A proofreader catches errors. A continuity manager tracks what was established where across thousands of pages. Brandon Sanderson's novel Wind and Truth was produced with the support of 1.3 million words of beta reader feedback, a dedicated continuity manager responsible for the series wiki, and a VP of editorial building production spreadsheets (Ahlstrom & Dragonsteel Books, 2025). The intellectual direction — what the story claims, how it argues, what it is for — remained his. The execution apparatus did not determine what the work was for.

This is not exceptional. It is the standard professional model. Nobody treats editorial collaboration as evidence against authorship. Even where editorial contribution is substantial, the convention is that intellectual direction remains with the named author — the collaborators serve the author's vision, not their own. The acknowledgments page lists them. The author's name goes on the cover.

What AI changed is not the logic of this arrangement but the resource requirement for accessing it. The delegation that was always philosophically compatible with authorship was, until recently, expensive and mostly invisible. Professional editorial infrastructure was gated by contracts, relationships, and cost. AI made the same fundamental arrangement — human direction, delegated execution — accessible to anyone with a browser, and made the execution layer scannable in a way it had never been before.

The Vending Machine Assumption applies a standard to AI-assisted writing that publishing has never applied to editorial collaboration. A writer working with AI assistance is flagged for delegating lower-level work. A writer working with a professional editing team is not. The underlying delegation is the same. What changed is who can afford it and whether a tool can see it.

Why the Pangram Score Answers the Wrong Question

> **Evidence**
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> PeerPrism separates idea and text provenance across 20,690 peer reviews. Under hybrid conditions, including human-origin evaluative ideas rendered in AI-generated text, detector predictions diverged from the intellectual source.

Pangram's own documentation shows what the chisel looks like from the inside.

What the detector reads

Pangram describes its own task precisely. From its knowledge hub: "Fundamentally, AI detection is a problem of author identification — categorizing which decisions are prototypical of what kind of author" (Pangram, n.d.a). That is a legitimate technical question. Its classifier learns patterns across a large training corpus and places new texts in learned representational regions. No single visible feature determines the result.

Separate research confirms that measurable linguistic regularities in LLM-generated prose are real. Reinhart and colleagues, publishing in the Proceedings of the National Academy of Sciences in 2025, found that instruction-tuned LLM prose contains consistent surface patterns: present participial clauses appear at 2–5 times human rates; GPT-4o used "camaraderie" 162 times and "tapestry" 155 times more frequently than humans in their corpus (Reinhart et al., 2025). These are lower-level execution properties — precisely the kind of signals Directed Authorship delegates. They tell you something about how the sentences were produced. They tell you nothing that settles the question of intellectual contribution.

What it gets asked to settle

The problem is not detection. It is what the detection is asked to settle. PeerPrism — a 2026 benchmark of 20,690 peer reviews presented at SIGIR — was explicitly designed to separate idea origin from text origin, including cases where human-origin ideas were rendered entirely in AI-assisted text. Its authors found that detector predictions diverge sharply under hybrid conditions and conclude that current detection methods conflate surface realization with intellectual contribution (Arvan et al., 2026). Once idea provenance and text provenance diverge, a detector of the second cannot settle the first.

The Chisel Fallacy occurs not inside the detector but one step later. Pangram may answer its own question correctly. That answer is still the wrong evidence for a different question: who supplied the upper-level intellectual contribution.

Two cases

In August 2026, UC Berkeley mathematics professor Zvezdelina Stankova published an op-ed arguing that student mathematical preparedness had declined following test-blind admissions. A student ran the piece through Pangram. It returned 33% AI-assisted. Stankova confirmed that AI had helped with editing and locating documents (Daily Californian, 2026; The Guardian, 2026). The response was immediate: the production process became the story.

Stankova's disclosed use is consistent with Directed Authorship: AI assisted with editing and document location — lower-level execution tasks. Her disclosed use

says nothing by itself about who supplied the diagnostic comparison or causal inference. Those are precisely the intellectual questions a detector cannot settle.

Stankova's argument is contestable on its own terms. The diagnostics she used changed across the periods she compared; causal attribution is genuinely difficult; her definitions of readiness can be scrutinized. Those are substantive questions worth engaging. Whether AI participated in the text, whether that participation was appropriately disclosed, and whether the argument is sound are three different questions. A detector bears directly on the first. It does not settle the second or third.

These are not the same question, and they require different kinds of evidence:

Question	**Right kind of evidence**
Was AI involved in the text?	Detector score
Was the involvement disclosed?	Disclosure norms / author statement
Is the argument sound?	Evaluating the argument itself
Who supplied the intellectual contribution?	Intellectual Work Ledger

The Chisel Fallacy is using evidence for the first question to settle the fourth. The IWL is worth its cost precisely when the fourth question is what is actually being asked.

The Commonwealth Short Story Prize encountered a version of the same problem. When AI allegations circulated against regional winner Jamir Nazir's story "The Serpent in the Grove," the Commonwealth Foundation did not use AI detection tools in its review. Its statement cited concerns about artistic ownership and consent surrounding unpublished work, while also noting that such tools may serve as useful indicators but "cannot provide conclusive evidence on their own" (Commonwealth Foundation, 2026; Brittle Paper, 2026). Instead the Foundation held detailed discussions with the regional winners and examined working drafts, timestamped documents, and notes before concluding that AI had not written the winning stories.

The Foundation behaved as though textual evidence alone were insufficient to distinguish delegation from authorship: it inspected the development record instead.

That move — from execution signal to development record — is the first half of this paper's argument in practice.

A perfect detector of AI involvement would still not be a detector of intellectual contribution.

When the Metric Becomes the Target

> **Evidence**
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> In a study of 642 published English abstracts, Pangram 3.2 flagged 64.4-80.1% of abstract-only refinements at a 0.50 threshold. After commercial humanization, over 96% of AI-labeled rewrites were reclassified.

> **Implication**
>
> When a production flag becomes a reason to stop reading, useful knowledge can be dismissed before its evidence or reasoning receives examination.

The relationship between policy targets and statistical measures is what Goodhart documented in 1975: once a measure is used as a policy target, it tends to lose the information value it had when observed independently (Goodhart, 1975). The mechanism is operating here.

A 2026 study from the University of Notre Dame, presented at the ACM AI Leadership Summit, tested a proxy for guideline-compliant AI assistance — light editing of the kind many institutional policies broadly permit — and submitted the results to Pangram and GPTZero. Pangram flagged 64–80% of these lightly edited pieces. Then the researchers took fully AI-generated text, ran it through a commercial humanizer, and resubmitted. More than 96% escaped detection. The authors' conclusion: "Honest AI-editing results in a higher sanction risk than humanizer-assisted evasion" (Karr et al., 2026).

The Market's Response

The market adapted almost immediately. A cottage industry of human specialists now exists on Fiverr whose entire value proposition is defeating the authentication system. One listing: "Manual Rewriting. No spinners or bots. Every word rewritten by me. Burstiness: Short and long sentences mixed naturally. AI Detector Proof. Tested before delivery. It will pass" (Fiverr, n.d.). The system ends up rewarding human execution introduced specifically to conceal the provenance it was built to detect.

When a detector score becomes a compliance target, the market optimizes against the score rather than the underlying intellectual behavior.

There is a consequence beyond Goodhart's Law: detector enforcement teaches authors that disclosure is the risky choice. Stankova disclosed her AI use and the disclosure became the story. A regime that flags honest editing at 64–80 percent while humanizer-assisted evasion passes at 96 percent is not incentivizing transparency — it is penalizing it.

The metric was also fragile before anyone tried to game it. The same study found that Pangram flag rates varied from 2.1% for chemistry abstracts to 26.6% for theology abstracts (Karr et al., 2026). Detector scores tracked stylistic features that are not unique to AI involvement. Once the score becomes consequential, the optimization pressure compounds an already-confounded measurement.

The result is a metric squeezed from both sides: natural confounding below, strategic evasion above. A high human score no longer uniquely identifies unaided authorship.

The Floor Is Already Moving

The same transition has run through other domains before it reached writing. In each, automation lowered an execution floor and the argument about what constitutes meaningful contribution shifted upward. The mathematics case shows the pattern most clearly — it has run for decades and is still accelerating.

Mathematics

For much of mathematical history, "computer" was a job title. Human beings performed arithmetic by hand. When calculators arrived, the floor migrated. Then symbolic computation software automated algebraic manipulation. Then formal proof assistants automated the logical verification of proof steps. The resistance followed a consistent pattern: the delegated layer was claimed as essential to genuine mathematical understanding. In each case, the work that came to be recognized as the point was located at the layer above.

AI is moving the floor again. Tao, writing on mathematics following the 2026 International Congress of Mathematicians, describes the emerging practice: humans author the statements of theorems — determining what to prove and why it matters — while automation handles the proofs (Tao, 2026). A system of 10,000 autonomous AI agents recently submitted a claimed solution to the Navier-Stokes blowup problem for formal verification, awaiting review by the Clay Mathematics Institute — a result that, if it stands, illustrates exactly where the floor is now moving (Quanta, 2026; Nature, 2026). The question that was always the hardest — what is worth knowing, and what does a result mean — remains with humans.

Programming

In February 2025, Karpathy named a shift already underway: vibe coding — "forget that the code even exists" (Karpathy, 2025). By November 2025, Collins Dictionary named it Word of the Year (Collins Dictionary, 2025). By April 2026, Karpathy described the pattern as having shifted to "agentic engineering" — more deliberate oversight, same fundamental structure: developer above the syntax floor, AI handling execution (Cook, 2026).

Programming increasingly accepts substantial AI-generated implementation as compatible with genuine professional contribution, while shifting evaluative weight toward specification, architecture, review, and orchestration. By early 2025, Microsoft was reporting 20–30 percent of its production code AI-generated, with Google's CEO reporting well over 30 percent at the same period — both presenting the figures as productivity metrics rather than authorship concerns (TechCrunch, 2025). By February 2026, Cisco's president was explicit: "They should master orchestration and innovation, not syntax. I would rather our people are thinking about the next big thing, not syntax" (Euronews, 2026).

Writing is navigating this same question, with a complication: writing has a tool — Pangram — that can see the execution layer, converting a debate that other fields worked through quietly into an institutional enforcement mechanism.

A verification regime that treats mastery of the old execution floor as a condition of legitimacy risks screening out intellectual contributions whose value is independent of that floor. The floor was always delegatable — professional editorial infrastructure proved it. What AI changed was who could afford it and whether a tool could see it.

The institutions that evaluate competence most directly are beginning to adapt. UC Berkeley declined to continue Turnitin's AI detector following a pilot spanning Fall 2023 through Spring 2025, citing unclear actionability (UC Berkeley RTL, n.d.). The University of Cape Town moved toward process-oriented assessment. Melbourne has shifted toward staged work and oral demonstration of understanding. These are not institutions that have solved the problem. They are institutions recognizing that surface detection cannot answer the question they actually need answered.

When the execution floor moves, evaluation has to move with it.

A Better Standard

> **Definition**
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> The Intellectual Work Ledger is a proposed framework linking a Development Ledger, Decision Record, Authorial Accountability, and Dependency/Proof Chain to make claims about intellectual contribution checkable.

> **Counterpoint**
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> A system can generate an article, intermediate drafts, rejected alternatives, and retrospective commentary together. Their agreement can reflect a coherent generated history rather than an independent record.

Process provenance and what it cannot do

But process evidence is only the beginning. The institutional moves described above represent a genuine shift toward inspecting development rather than merely judging finished output. Working drafts, version histories, timestamps, and oral defenses can establish that a creation process occurred. They answer questions about process provenance: what happened, in what sequence, and through whose actions. But they do not necessarily answer the question this paper is most concerned with: does the claimant command the intellectual structure the artifact claims to represent?

Emerging technical work already formalizes the first problem. A 2026 individual Internet-Draft titled Proof of Process proposed a cryptographic framework for making a continuous human authoring process tamper-evident through behavioral, temporal, and chained evidence (Condrey, 2026).[1] That is useful process provenance. It does not by itself establish why the argument has the structure it does, which alternatives were considered and rejected, what evidence bears on which claim, or what would happen to the conclusion if one of its premises failed.

Process timestamps and the IWL address different axes and are stronger together: a timestamped development record anchors IWL entries in time, forcing a fabricator to construct plausible reasoning in real time rather than retrospectively — each axis closes gaps the other leaves open.

That is a different axis of evidence.

Most proposals for improving authorship verification try to detect the counterfeit better. The IWL is structurally different: it raises the cost of counterfeiting until the counterfeit approaches the cost of the genuine article.

The Intellectual Work Ledger

The principle behind the IWL is one most people encountered in mathematics class: show your work. Not just the answer — the path. The reason teachers required this was diagnostic. If you derived the answer yourself, showing the path is easy — you are describing something that happened. If you copied the answer, reconstructing a believable path is hard. You have to reverse-engineer a journey you never took.

Something like this already exists in intellectual work, informally. Research papers trace every claim to a source. Investigative journalism maintains source documentation. Fact-checkers preserve the chain from conclusion to evidence. Grant applications require methodology sections. These practices exist because intellectual claims are not self-certifying — they need to be traceable. The IWL formalizes the same principle and applies it to a question those practices do not fully address: not

just where the facts came from, but whether the author commanded the reasoning that connects them.

The IWL is designed to exploit an asymmetry: genuine work can often be documented from decisions that already occurred, whereas fabrication must retrospectively construct and reconcile those decisions. That asymmetry is a design property of the IWL, not a proven theorem — its strength depends on the depth and independence of the constraints applied.

The Intellectual Work Ledger does not attempt to prove that a human typed the artifact. It makes the intellectual work behind the artifact inspectable.

The IWL is not four documents demonstrating that someone appeared to think. It is a multi-surface constraint architecture. A Development Ledger preserves the meaningful evolution of the work: discoveries, corrections, reversals, and the sequence in which later decisions became possible. A Decision Record preserves alternatives, rejections, and the reasons choices were made, exposing the evaluative landscape behind the final artifact. Authorial Accountability identifies the claimant who assumes responsibility for the reasoning and can be challenged on it. A Dependency and Proof Chain exposes how evidence, premises, decisions, and conclusions depend on one another.

Each surface is useful independently. Together they constrain one another. A later decision cannot be freely rewritten if earlier evidence already constrained it. A rejected alternative has to make sense given what was known at the time. A claimed source must actually bear on the conclusion attributed to it. The IWL is designed to surface what the Vending Machine Assumption obscures — the upper-level intellectual work that may or may not be present behind any AI-assisted artifact.

The IWL is not only a defense for authors practicing Directed Authorship. A one-shot vending-machine user who performed no intellectual work on the artifact cannot produce a genuine Development Ledger, cannot reconstruct actual decision points and their reasoning, and cannot survive unpredictable challenge on the intellectual process — because there was none. The IWL settles the authorship question for both workflows. It replaces detection not by being more sympathetic to AI-assisted authors but by asking a question that the right answer can actually answer.

If you actually did the intellectual work, the IWL is largely a documentation task. You are recording decisions that already happened — what you found, why you changed direction, which source actually supported the claim you made from it. The record is straightforward because you are describing something that occurred.

If you did not, the record becomes the work — you must construct, retrospectively, a record of reasoning you now have to perform rather than describe.

Consider a deliberately hypothetical fabricated entry — not a claim about Stankova's actual process: "Selected the 2019 diagnostic instrument rather than course grades as the readiness measure because the diagnostic is domain-specific to calculus readiness, while course grades reflect effort and attendance effects that contaminate the preparedness signal." That entry sounds authoritative. Now the IWL's second level of constraint contact applies: does a 2019 diagnostic instrument with these properties actually exist? The fabricator looks. Does it match what Berkeley's calculus sequence used? They check. Is the contamination argument actually supported in the measurement literature, or is it an assumption introduced to sound rigorous? They search.

Three questions. One entry. To make a single plausible Decision Record hold up, the fabricator has now engaged with mathematics education diagnostics, Berkeley's institutional assessment history, and the measurement literature on grade contamination. A fabricator capable of answering all of those questions consistently has, in the process, performed something that closely resembles the intellectual work the entry claims to document. Multiply that across a complete IWL and the fabrication cost structure becomes visible.

Recent philosophical work on understanding helps explain why. Queloz and Beckmann argue that understanding is not merely possession of propositions or successful reproduction of surface patterns; it involves a model of the relationships within a domain robust enough to support prediction, explanation, and competent response beyond memorized cases (Queloz & Beckmann, 2026).[2] Intellectual command is relational: knowing what supports what, what depends on what, why one alternative was rejected, and how changing one premise propagates through the rest of the structure.

The IWL is designed to make portions of that relational structure inspectable.

Verification operates at two levels. The first asks whether the pieces exist: does the source exist, is the timestamp genuine? The second asks whether the claimed relationships hold: does the source actually support the claim, does the evidence justify the rejection, does the premise generate the conclusion? The second level is what ordinary provenance records cannot provide. Verify constraint contact, not narrative plausibility. A coherent story can be fabricated. The harder target is a relational structure that must remain consistent with independently anchored evidence and survive challenges that were not known when the story was produced.

The Fabrication Paradox

The security property the IWL exploits has a name: costly signaling. Michael Spence demonstrated in 1973 that a signal is credible when it is cheaper for the type with the genuine property to produce it than for the type faking it — education works as a signal of ability because capable candidates bear lower costs com

pleting it than incapable ones faking their way through (Spence, 1973). Zahavi's handicap principle applies the same logic in biology: the peacock's tail is an honest signal of health because only a genuinely healthy bird can afford to grow it (Zahavi, 1975). The signal is honest because the cost runs in the right direction. The IWL is the same machinery applied to authorship: if you did the intellectual work, assembling the trail is nearly free — you are describing what happened. If you did not, fabricating a plausible intellectual journey backwards — the dead ends, the revisions, the order things were figured out, the pressure tests that forced definition changes — approaches the cost of having the journey.

What if the only reliable way to fake a record of intellectual work was to do the intellectual work?

That is not a rhetorical question. It describes a real property a verification system can approach — one where the cheapest path through fabrication increasingly resembles performing the thing the fabricator was trying to avoid. This paper proposes calling it the Fabrication Paradox.

A generative system can cheaply produce a ledger, a decision history, rejected alternatives, and a proof chain. That is not the objection to overcome. It is the starting condition. The question is what happens as the verification gets more demanding.

For that fabricated record to become convincing evidence of intellectual provenance, it must remain consistent with the final claims, external evidence, chronology, decisions, alternatives, dependencies, and the claimant's responses under unpredictable scrutiny.

Two costs then begin to rise together. The first is a fabrication-cost gradient: as independently anchored constraints accumulate, the space of plausible false histories narrows. The second is an epistemic-cost gradient: establishing that the fabricated history deserves confidence increasingly requires reconstructing why the argument has the structure it does. The fabricator must recover not merely the nodes but the evidential, dependency, counterfactual, and evaluative relationships among them.

These gradients couple: as the space of plausible false histories narrows, navigating it requires reconstructing more of the relational structure the genuine work produced — which is precisely the epistemic work the fabricator was trying to avoid.

This is the consistency tax: the fabricator must satisfy cross-channel consistency simultaneously — timeline, diffs, decisions, dependencies, and real-time responses all constraining one another — and the cost is multiplicative rather than additive. Reality gives the honest researcher this consistency for free. Simulation requires reconstructing it.

Recovering those relationships is itself epistemic work. The better the counterfeit of intellectual competence must become, the more the counterfeiter must acquire the competence being counterfeited.

This is a proposed mechanism, not a theorem that fabrication is impossible or that fraudulent production must always cost more than honest work. The narrower claim is that stronger cross-consistency requirements and less predictable challenges progressively remove the cheapest forms of fabrication by forcing the counterfeit to reproduce more of the relational structure that gives genuine intellectual work its coherence.

A fabricator who delegates the verification step faces a dilemma. If they trust the delegated check without personally confirming the ledger holds together, they cannot tell a sound record from a subtly broken one — and they are exposed the moment a challenge arrives that the delegated checker did not anticipate. If they actually confirm the ledger makes sense, they have performed the intellectual work the ledger claims to document. The only escape from both horns is to never be personally questioned. Authorial Accountability — the layer of the IWL that requires the named claimant to stand behind the reasoning under unpredictable scrutiny — is what forecloses that escape. Delegation of fabrication and delegation of verification both terminate at the point where a human has to answer.

This includes AI-assisted fabrication: a system can generate constraint-satisfying ledger entries, but the named claimant still has to answer for them under scrutiny that is unpredictable in real time. The AI cannot anticipate what will be asked. The claimant, not the system, bears the accountability.

Cryptographic proofs of knowledge provide the relevant analogy: producing a convincing transcript of expertise is not the same as possessing it — appropriately varied challenges are what make genuine possession inferable (Bellare & Goldreich, 1992).[3] Bitcoin's proof-of-work supplies a related intuition: verification can be designed so that dishonest participation becomes progressively more costly (Nakamoto, 2008). An IWL borrows from both: challenge-based evidence of command from proof-of-knowledge, and adversarial cost-raising from proof-of-work.

None of this eliminates the verification boundary. A sophisticated fabrication may survive. A coherent ledger cannot prove the exact private mental history of its claimant. The aim is narrower and more useful: raise the boundary. Make claims answer to independent evidence. Make decisions answer to alternatives. Make conclusions answer to their premises. Make the claimant answer to the structure that supposedly produced the work.

The bar is economic, not cryptographic: the IWL deters the realistic threat — output passed off as sustained intellectual work — not an adversary with unlimited resources and no accountability requirement.

The IWL does not make intellectual provenance certain. It makes it increasingly inspectable — and increasingly difficult to counterfeit without reconstructing the intellectual work itself.

This Paper's Own IWL

A standard for intellectual provenance should apply to the paper that proposes it. The excerpt below shows what that looks like in practice — not a summary of what happened, but a sample of the actual record, with the fabrication cost made explicit for each entry.

Development Ledger — Selected Entries

Phase	Event	What changed	Why it mattered
v2 phase	VMA coined — missing causal antecedent identified	v1 distinguished two workflows but could not explain why critics feel detection is sufficient; Vending Machine Assumption named the antecedent	Closed the logical gap; made the Chisel Fallacy structurally inevitable rather than arbitrary
v6 phase	Agency Gradient coined — upstream variable identified	AI-assisted work identified as spanning a broad spectrum of retained human agency; named as the empirical fact the VMA was implicitly ignoring; identified as path to a future paper not yet published	Named the missing upstream variable; VMA now framed as an error about the gradient rather than a free-standing belief

Entry 1 — VMA coined

> ***Author:*** *v1 exists as a discrete draft. This entry describes a reading event — noticing a specific gap in the argument — and the observation that named it. Near-zero documentation cost.*
>
> ***Fabricator:*** *Must construct a complete ~25-page v1 containing the full argument structure, all cases, all sources — but specifically and consistently missing the VMA throughout. Must produce a plausible account of the naming event. Must anchor this in inspectable document history that a reader could verify. Cost: producing the paper's entire prior state.*

Entry 2 — Agency Gradient coined

> ***Author:*** *Insight arrived during a review pass of a near-complete paper. Entry describes where it came from and what it opened up.*
>
> ***Fabricator:*** *Must explain why this concept arrived at v6 specifically — after the paper was considered structurally complete — rather than v1 or v3. Must*

> *account for the "future paper" observation, which requires knowing what the next investigation will be about. That information is not in the published paper.*

Decision Record — Selected Entries

Decision	Rejected	Adopted	Reason
Chisel Fallacy location	Pangram itself commits the fallacy	Fallacy occurs in downstream inference from Pangram's score	If the fallacy requires Pangram to be wrong, any technical improvement to Pangram weakens the paper. Moving the error downstream means Pangram can be entirely correct and the Chisel Fallacy still operates. Specific architectural judgment about the paper's own vulnerability.
DA definition scope	"A mode of AI-assisted production in which the author delegates lower-level execution"	"Any mode of production in which intellectual direction is retained by the author while lower-level execution is delegated to any capable agent, human or otherwise"	The paper's primary illustration of DA is Brandon Sanderson — no AI involvement. The AI-specific definition excluded its own most important example. Pressure test: does the current definition include what Sanderson is doing? No. The concept was right, the words were wrong.
Domain example quantity	Six domain examples: programming, mathematics, music, Yates, Xania Monet, Karpathy	Two primary cases: mathematics and programming	Each case should add a new constraint to the argument — a different mechanism or type of evidence — not restate the thesis in a new domain. Quantity was diluting rather than strengthening. The paper's own argumentative standard applied to its own structure.

Entry 1 — Chisel Fallacy location

> *The reason column states the architectural judgment. The entry is its own demonstration.*

Entry 2 — DA broadened to tool-agnostic

> **Discovery Context / Authorial Rationale — DA definition scope:** *Throughout the earlier drafts, DA had been implied rather than clearly stated — a concept I was actively engaging in, but the definition had not caught up with what I actually meant. The AI-specific framing had crept in because the paper is nominally about AI writing, but DA as I understood it was never AI-specific. The definition clarification came from a simple pressure test. I looked at Sanderson and asked: does our current definition include what he is doing? The answer was no — "AI-assisted production" excluded him by definition. That was the problem. The concept was right but the words were wrong. Once that was clear, the definition wrote itself.*
>
> ***Author:*** *The AI-specific definition exists in v1–v3. Entry describes a pressure test and the criterion it applied. The Discovery Context note above describes the moment the revision became clear.*
>
> ***Fabricator:*** *Must produce five or six prior versions where DA is consistently AI-specific — while Sanderson appears as the primary publishing example in Section 5 throughout those same versions. Must hold that internal inconsistency deliberately across a version history. Must construct a Discovery Context note with the specific register of an instant recognition from a specific trigger, then ensure every anchor it contains checks out against the published paper.*

Entry 3 — Domain examples six → two

> ***Author:*** *The structural problem was visible in a v3 read-through. Entry records the criterion used to resolve it.*
>
> ***Fabricator:*** *Must know which six examples existed in v3 by name. Must know the dilution was diagnosed as a constraint issue specifically, not a length or word-count issue. Must know the criterion — "each case adds a new constraint" — which is stated nowhere in the published paper.*

The entries above demonstrate the Fabrication Paradox concretely rather than in principle. Each contains information that is nowhere in the published artifact — what v1 lacked, which six examples v3 had by name, why the DA definition changed when it did and what specific example forced it. A fabricator working from the published paper alone is stranded. The real author has all of it because it is simply what happened. The entries are easy to produce if you were there and hard to produce if you weren't — not because of their length but because of what they contain. That is the IWL's actual verification property.

The fabrication cost is not uniform. The VMA entry requires producing the paper's entire prior state. The DA entry requires holding an internal inconsistency

across multiple version states and constructing a Discovery Context note whose register and specific trigger are hard to engineer deliberately. The six-to-two entry requires knowing a specific list and a specific criterion that appears nowhere in the published paper. Different entries surface different kinds of constraint. The consistency tax is visible across all of them: the fabricator must maintain consistency simultaneously across version states, decisions, Discovery Context notes, and responses under challenge — and that cost is multiplicative, not additive.

The paper is its own test case, not its own proof. The complete Intellectual Work Ledger is published alongside this paper. Whether the entries hold up under scrutiny is the test this section is proposing.

Future Directions

Two directions emerge from this paper. The first is the Agency Gradient as its own investigation: AI is a fundamentally open-ended tool with no social agreement on what "using AI" means, and that openness — the fact that the same model produces incomparable outputs depending on who operates it — is the upstream condition the Vending Machine Assumption has been collapsing. The second is ReTI (Researcher Timeline): a private-by-default researcher feed where a researcher commits moments of insight, source discoveries, and draft revisions as they happen — hash-chained, notary-attested, reversals included. The consistency tax observation suggests a design rule for ReTI: every new temporal channel should cost the honest researcher nothing and the fabricator compounding amounts. Timestamps are free for you, expensive to fake. Diffs are free for you, expensive to fake. ReTI is the temporal implementation of the same principle the IWL applies to content — and the unflattering commits you would delete are the strongest authenticity signals, because a fabricator writes a highlight reel. Both directions were created by the work this paper did.

The Chisel and the Sculptor

The detector flags the chisel. It is completely blind to the sculptor.

We almost never read to evaluate syntax. We read to encounter ideas, to test arguments, to follow reasoning to its conclusion.

The calculator does the arithmetic. The mathematician does the math. The developer directs the architecture. The AI writes the syntax. These separations have been contested in domain after domain as each execution floor became automatable. The debate is familiar: the delegated layer is claimed as essential to genuine understanding; the work above it is eventually recognized as the point. Writing is

navigating this now, with a complication that other fields did not have — a tool that can see the execution layer, making a quiet debate into an enforcement mechanism.

Execution-layer evidence cannot by itself settle whether intellectual contribution was present. A detector that identifies AI involvement in syntax production has identified exactly what it was built to identify. What it cannot recover is whether the author retained intellectual responsibility for what the artifact claims and why. That is a different question. It requires different evidence.

Pangram is a very sophisticated way of asking the wrong question.

Notes

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